

FOREIGN BODIES IN THE AIRWAY

<u>Presentation</u>	
Asymptomatic Witnessed complaint, history of recent choking Fussiness, inconsolable crying	
Unilateral hyperinflation on CXR, recurrent pneumonia or atelectasis on the same side	Suggests airway obstruction from FB
Respiratory: stridor, unilateral wheeze, resp distress	Suggests airway obstruction from FB

Recognition of foreign body causing airway obstruction is crucial, especially when there are signs of severe airway obstruction. This includes a silent cough, increasing respiratory difficulty, stridor, or patient becoming unresponsive.

In an emergency like this, the immediate management would be to:

1. Call for help.
2. Try to dislodge object with 5 back blows and then 5 chest/abdominal thrusts**

-back blows – support the child by placing 1 arm diagonally across child's chest and lean patient forward, then firmly strike child between shoulder blades with heel of your hand with 5 of these back blows.

AND

-chest thrusts (for <1 year old) – Turn the infant supine, place infant on your thigh and support infant's occiput and upper back with your hand. Identify landmark for chest thrusts (same as that in chest compression - lower sternum, 1 finger's breadth above xiphisternum). Deliver 5 chest thrusts (sharper in nature, slower compared to chest compression).

or

-abdominal thrusts (for >1 year old) – Stand behind child and wrap arms around child's waist. Locate child's belly button and make a fist with the thumb side against the middle of the child's abdomen just above the navel and well below the lower tip of child's breastbone. Give 5 quick, upward thrusts into the abdomen.

3. Continue alternating 5 back blows and 5 chest/abdominal thrusts until the object is forced out or if patient becomes unconscious. Start CPR immediately if patient becomes unconscious.

**chest thrusts for infants (ie < 1 year old), abdominal thrusts for > 1 year old

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If there is partial airway obstruction (ie able to breathe, cough effectively and cry/speak), encourage child to cough and monitor continuously. If the coughing is/becomes ineffective, call for help and start delivering back blows and chest/abdominal thrusts (as above), and repeat the sequence until the obstruction is relieved or the patient becomes unconscious. Start CPR if patient becomes unconscious at any point.

Consider lateral neck and CXR if patient is stable.

NASAL FOREIGN BODIES

1. Presentation:

Most nasal FBs are located on the floor of the nasal passage just under the inferior turbinate or superiorly in the nasal cavity just in front of the middle turbinate.

Most children present with a history of nasal FB insertion without symptoms; while some present with nasal discharge, epistaxis, foul odour or nasal obstruction. Consider facial X-ray to confirm position of nasal FB.

2. Instruments to use (see Figure 1):

a) For graspable FB: Use a small Alligator forceps or Hartmann forceps (not a Tilley forceps as it is too big for most patients).

b) For smooth and non-graspable FBs eg: beads, stones: Use a blunt right-angle hook.

Figure 1: Instruments to use



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- c) Suction can be useful to clear mucus, and on occasion, it can help remove small nasal FBs.
- d) Ask patient to blow nose with empty nostril occluded; or try PPV to mouth with normal nostril compressed.

3. Technique for removing an intra-nasal FB:

- a) Ensure adequate lighting via a headlight or lamp for good visualization.
- b) If patient is cooperative and FB is readily seen in the anterior nasal cavity, sit patient upright and use the non-dominant hand to push the tip of nose up.
- c) If patient is uncooperative, proper restraints are essential and consider sedation.
- d) Use appropriate instruments for removing FB (this depends on type of FB); for deeper FBs, insert a nasal speculum for better visualization.

4. Urgent ENT referral:

Nasal FB extraction is usually an elective procedure except for:

- a) **Button batteries**
- b) **Paired disc magnets**

which can cause serious damage to nasal structures and warrant urgent removal.

5. ENT referral is warranted for:

- a) All button batteries or magnet FB in nose.
- b) Posterior FBs (not readily visualized by anterior rhinoscopy).
- c) Chronic or impacted FBs associated with marked inflammation.
- d) Penetrating or hooked FBs.
- e) Any FBs that cannot be removed at initial attempt due to poor cooperation, bleeding or limited instrumentation.

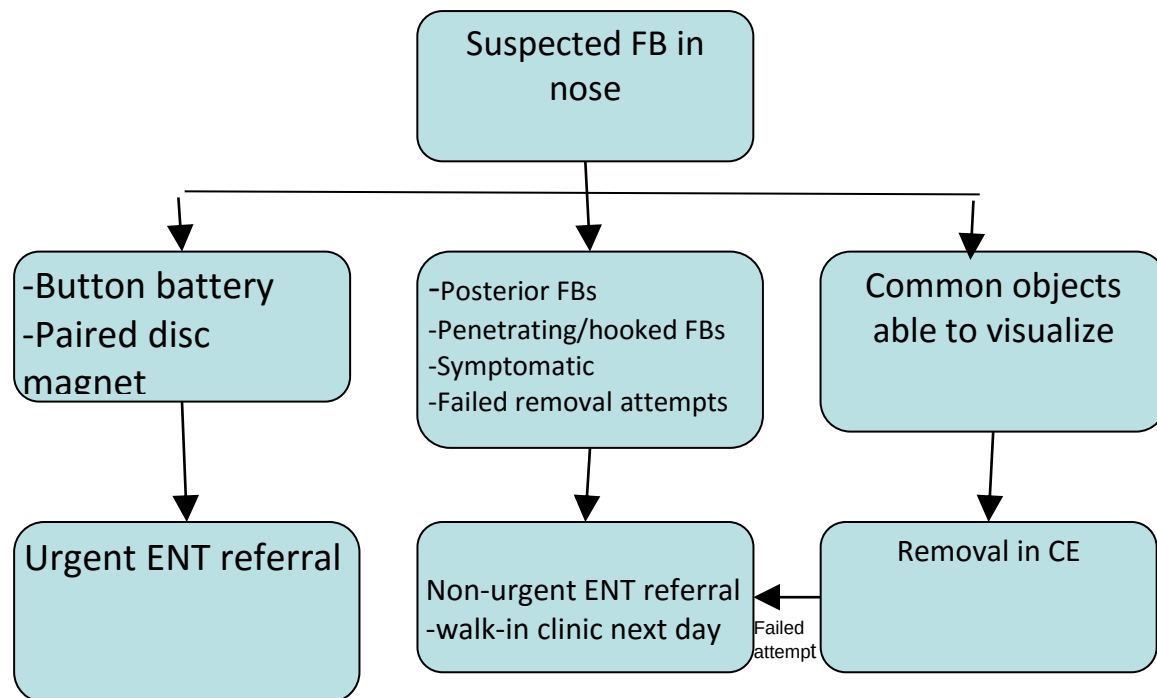
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Figure 2: Workflow for intranasal FB



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EAR FOREIGN BODIES

1. Use proper headlight and appropriate sized aural speculum.
2. Attempt to remove ear FB only once
3. Instrument of choice depends on ear FB:
 - a) For graspable ear FBs – use alligator forceps or a small 90 degree hook for bigger items.
 - b) For non-graspable ear FBs – try suctioning and flushing to remove FBs. However, do not flush organic materials or seeds as these FBs can expand. If unable to flush out FB, attempt removal using a cerumen curette.
 - c) For light material - try suctioning
 - d) Other instruments: Irrigation setup, headlight, otoscope, ear speculum
 - e) Irrigation method: Before irrigation, make sure TM is intact and no grommets present. Place patient in supine position with affected ear up; instill mineral oil or 1% lignocaine to kill if insect, then place towel/basin under patient's head to catch irrigant solution as it flows out of ear. Direct irrigation stream along posterior and superior margin of EAC and briskly depress the syringe until FB is expelled.
4. Location of FB (see Figure 3):
 - Ear FB in the outer third – can be safely removed without sedation in a cooperative patient
 - Ear FB in the middle third – attempt once without sedation and if failed or FB pushed to inner third, abort procedure. Refer ENT clinic on the next working day. If sedation is planned, call ENT on call.
 - Ear FB in the inner third – remove under sedation. If non-urgent (eg: no ear pain, no infection), refer ENT next working day. Otherwise refer ENT on call.

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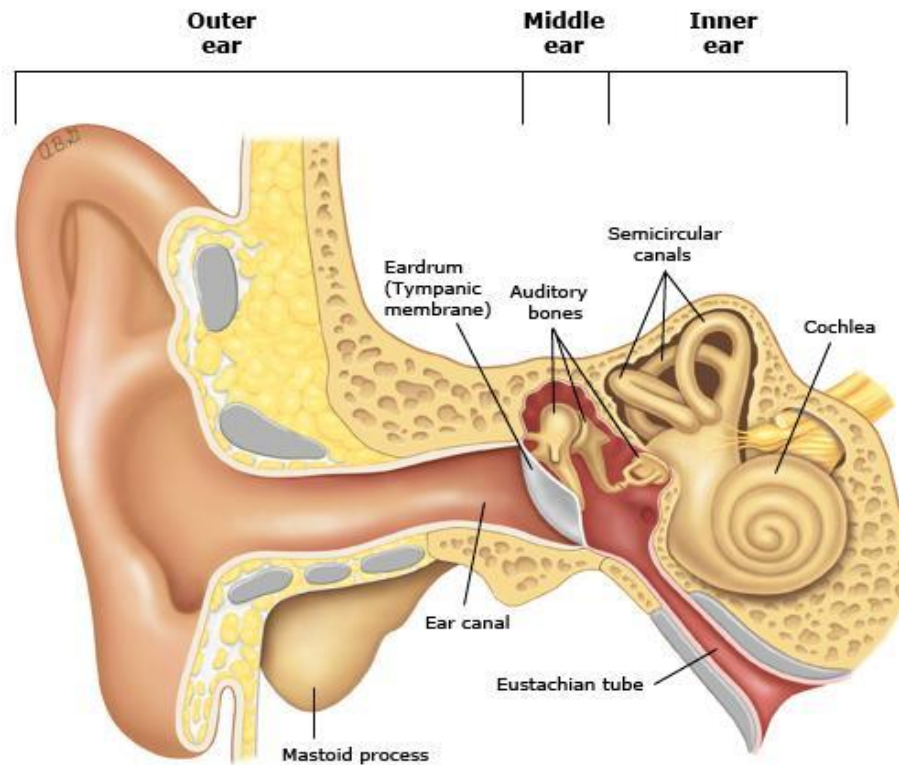


Figure 3: Normal structures of outer, middle and inner ear

5. Type of FB determines timing for removal:

-Button batteries, live insects, penetrating foreign bodies warrant **urgent** removal

*button batteries: can destroy EAC and cause pressure necrosis

*insects: instill olive oil/ETOH or 1% lignocaine before attempted removal

*penetrating FB: can penetrate TM and cause damage to middle ear structures (eg: cotton buds, pencil points, hair pins)

-All other FBs (round beads etc) – elective removal if patient is asymptomatic

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6. Complication – EAC abrasion or laceration is the most common complication of FB removal and occurs in up to 50% patients. Risk increases with multiple attempts. Rx: ofloxacin ear drops. If canal is already swollen or traumatized from repeated attempts to remove FB, patient can be started on oral antibiotics and pain relievers. Do not give eardrops if there is a suspected TM perforation. Ear FBs are best removed after swelling has settled. Also advise parents to keep the affected ear dry.

7. When to refer ENT:

- button battery
- potentially penetrating FB
- live insects
- FB with evidence of injury to EAC/TM or middle ear as indicated by otorrhoea, vestibular symptoms (nausea/vomiting, nystagmus, vertigo, ataxia) or marked pain

Caveats - If patient has a URTI and requires sedation for FB removal, weigh benefits and risks of sedation. Discuss with senior on shift.

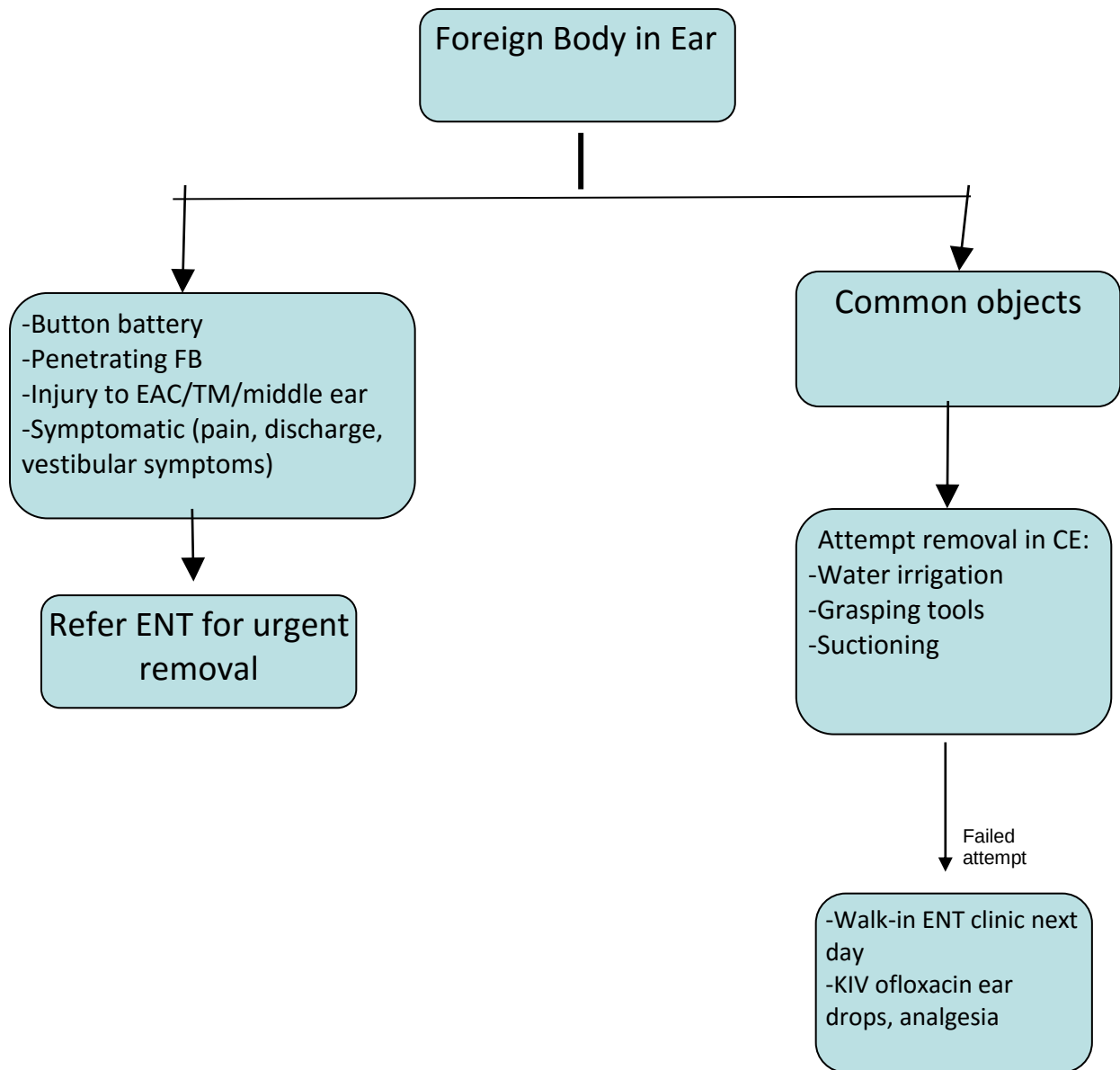
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Figure 4: Workflow for intra-aural FB



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PHARYNGEAL FBS

1. Ask the patient to pinpoint the exact location of discomfort or pain.
 2. Perform tracheal rock and check if patient can swallow a cup of water
 3. Perform a lateral neck X-ray in all patients.
 4. Decide if the pain is above or below the larynx. View the pharynx with a tongue depressor (and Tilley's forceps ready) . Visualize FB on throat examination as well as on direct laryngoscopy (if child is cooperative and physician is confident in performing this)– most throat FBs (especially fishbone) are lodged in tonsils, especially the inferior pole.
 5. If FB is not visualized on throat examination as well as direct laryngoscopy, refer ENT for visualization KIV removal under flexible nasopharyngoscopy.
 6. Keep patient fasted for sedation, counsel parents for sedation including procedure, cost and potential risks involved.
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7. Post-FB removal – discharge advice must be given:
Return to CE if increasing pain, hemoptysis, hematemesis or migration of pain to chest.
Keep TCU with ENT.

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